

# TinyBolus Formulary

Peer-review document

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Source: [github.com/SGP-Prime/tinybolusformulas](https://github.com/SGP-Prime/tinybolusformulas)

Privacy policy: [sgp-prime.github.io/tinybolus-site/privacy.html](https://sgp-prime.github.io/tinybolus-site/privacy.html)

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*This is a reference tool, not a prescribing system. Verify every value against your hospital protocols, product labelling, and the individual patient in front of you before administration.*

# About this document

This is the human-readable export of the TinyBolos paediatric reference formulary, organised section-by-section as it appears in the application. Each item is presented with its dose specification, applicable age brackets (where present), route, clinical notes, and numerical references pointing into the bibliography at the end of the document.

## How to read each item

Items fall into a small number of shapes that map directly onto how the application computes them:

- **Single dose** (e.g., 10 µg/kg): one factor multiplied by weight, often with a hard maximum cap.
- **Dose range** (e.g., 2–3 mg/kg): a low–high range, also weight-scaled, optionally with a maximum cap on either bound.
- **Age-bracketed**: different dose factors for different age cohorts. Rendered as a small table per item.
- **Age-bracketed fixed value**: not weight-scaled, just a string per age band (e.g., laryngoscope blade size).
- **Computed**: piecewise logic that doesn't reduce to a single formula (ETT sizing, fluid maintenance, LMA / supraglottic airway sizing, vital-sign normal ranges).

## Hard floors / contraindications

Some items carry an *age* or *weight* floor below which the formulary explicitly declines to suggest a dose. In the app this renders as ∅ (empty-set glyph) with a "not indicated" explanation. Items with such floors are flagged in red below each row.

## References

Every dose / range / age bracket / sizing rule carries one or more citations. References are numbered in encounter order throughout this document, with a full bibliography at the end. The numbers in [brackets] after each item refer to entries in that bibliography.

## Source

The formulary itself lives at [github.com/SGP-Prime/tinybolusformulas](https://github.com/SGP-Prime/tinybolusformulas) as a single machine-readable JSON file. This document is regenerated from that file on demand by `tool/generate_formulary_review_html.dart`; do not hand-edit it.

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# EMERGENCY — PALS

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## Compression rate

VALUE 100–120/min

References: [\[1\]](#)

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## Compression depth

RULE 4 cm <1 y · 5 cm ≥1 y

*not exceeding 6 cm at any age*

References: [\[1\]](#)

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## C:V ratio

VALUE 15:2

*15:2 if PBLIS-trained · 30:2 untrained*

References: [\[1\]](#)

---

## Defibrillation

DOSE 4 J/kg

*stepwise to 8 J/kg for refractory VF / pulseless VT (6th shock on) · adolescent 120–200 J*

References: [\[1\]](#), [\[2\]](#)

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## Cardioversion (synchronised)

DOSE 1 J/kg

*double per attempt to max 4 J/kg*

References: [\[1\]](#)

---

## Adrenaline (cardiac arrest)

IV/IO

DOSE 10 µg/kg (max 1 mg)

*repeat every 3–5 min · max 1 mg/dose*

References: [\[1\]](#), [\[3\]](#)

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## Amiodarone

IV/IO

DOSE 5 mg/kg (max 300 mg)

*after 3rd shock · repeat after 5th · max 150 mg subsequent*

References: [\[1\]](#)

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## Atropine

IV

DOSE 20 µg/kg (max 500 µg)

References: [\[1\]](#), [\[4\]](#), [\[5\]](#)

---

## Adrenaline (bradycardia)

IV/IO

DOSE 1–2 µg/kg

*unresponsive to oxygenation/ventilation · atropine if vagal*

References: [\[1\]](#), [\[4\]](#)

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## Adenosine (1st dose)

IV

DOSE 0.1 mg/kg (max 6 mg)

References: [\[1\]](#)

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## Adenosine (2nd dose)

IV

DOSE 0.3 mg/kg (max 18 mg)

*escalate 0.05–0.1 mg/kg steps to max 0.5 mg/kg*

References: [\[1\]](#)

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## Magnesium sulphate (torsades)

IV/IO

DOSE 50 mg/kg (max 2 g)

References: [\[1\]](#)

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# Vital Signs

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## Heart Rate

|         |             |
|---------|-------------|
| ≤1 mo   | 120–170 bpm |
| 2–12 mo | 100–160 bpm |
| 1–2 y   | 98–140 bpm  |
| 3–5 y   | 80–120 bpm  |
| 6–11 y  | 75–118 bpm  |
| ≥12 y   | 60–100 bpm  |

References: [\[6\]](#)

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## Systolic BP

|         |              |
|---------|--------------|
| ≤12 mo  | 72–104 mmHg  |
| 1–2 y   | 86–106 mmHg  |
| 3–5 y   | 89–112 mmHg  |
| 6–9 y   | 97–115 mmHg  |
| 10–11 y | 102–120 mmHg |
| 12–15 y | 110–131 mmHg |
| ≥16 y   | 90–140 mmHg  |

References: [\[6\]](#)

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## Diastolic BP

|         |            |
|---------|------------|
| ≤12 mo  | 37–56 mmHg |
| 1–2 y   | 42–63 mmHg |
| 3–5 y   | 46–72 mmHg |
| 6–9 y   | 57–76 mmHg |
| 10–11 y | 61–80 mmHg |
| 12–15 y | 64–83 mmHg |
| ≥16 y   | 60–90 mmHg |

References: [\[6\]](#)

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## Respiratory Rate

≤1 mo     40–60 /min

2–12 mo   30–53 /min

1–3 y     22–37 /min

4–5 y     20–28 /min

6–12 y    18–25 /min

≥13 y     12–20 /min

References: [\[6\]](#)

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## Tidal Volume

DOSE   5–7 mL/kg

*per IDEAL body weight — if obese use estimated weight · lower end for neonates*

References: [\[7\]](#), [\[8\]](#)

# Induction

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## Propofol

IV

≤1 mo 1–2.5 mg/kg

2–12 mo 3–4 mg/kg

1–6 y 3–3.5 mg/kg

7–12 y 2.5–3 mg/kg

≥13 y 2–2.5 mg/kg

*≈ halve after sevoflurane induction*

References: [\[9\]](#), [\[10\]](#), [\[11\]](#), [\[12\]](#)

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## Ketamine

IV

DOSE 1–2 mg/kg

References: [\[13\]](#), [\[14\]](#)

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## Ketamine (IM)

IM

DOSE 4–5 mg/kg

*when IV access impractical · onset 3–5 min*

References: [\[13\]](#), [\[14\]](#)

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## Rocuronium

IV

DOSE 0.6–1.2 mg/kg

*upper range for RSI*

References: [\[15\]](#), [\[16\]](#)

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## Succinylcholine (RSI)

IV

≤12 mo 2–3 mg/kg — *raised in infants — larger extracellular volume*

≥1 y 1–2 mg/kg

*intubating dose · avoid in MH / hyperkalaemia risk*

References: [\[16\]](#), [\[6\]](#), [\[17\]](#)



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## Fentanyl (induction)

IV

≤1 mo 1–3 µg/kg — *neonate: reduce · inject slowly (rigidity) · watch apnoea*

≥2 mo 1–3 µg/kg

References: [\[14\]](#), [\[18\]](#)

---

## Midazolam

IV

≤6 mo 0.1 mg/kg — *<6 mo: prolonged effect · slow injection · hypotension with opioids*

≥7 mo 0.1 mg/kg

CAP max 5 mg

References: [\[14\]](#)

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## Midazolam (oral premed)

PO

DOSE 0.5 mg/kg (max 20 mg)

References: [\[14\]](#)

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## Midazolam (buccal premed)

BUCCAL

DOSE 0.3 mg/kg (max 10 mg)

*if oral refused · seizures: see Emergency*

References: [\[14\]](#)

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## Midazolam (IN premed)

IN

DOSE 0.2–0.3 mg/kg (max 10 mg)

*5 mg/mL — divide between nostrils*

References: [\[19\]](#), [\[14\]](#)

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## Ondansetron

IV

DOSE 0.1 mg/kg (max 4 mg)

**Contraindicated <1 months**

References: [\[20\]](#), [\[14\]](#), [\[21\]](#)

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**Dexamethasone (PONV)**

IV

DOSE 0.15–0.25 mg/kg (max 4 mg)

References: [\[20\]](#), [\[22\]](#), [\[14\]](#)

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**Dexamethasone (anti-inflammatory)**

IV

DOSE 0.5 mg/kg (max 8 mg)

*airway oedema / post-extubation stridor*References: [\[23\]](#), [\[24\]](#)

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**Etomidate**

IV

DOSE 0.2–0.3 mg/kg

*for haemodynamic stability*References: [\[14\]](#), [\[6\]](#)

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**Atracurium**

IV

≤12 mo 0.3–0.4 mg/kg

≥1 y 0.4–0.5 mg/kg

References: [\[14\]](#), [\[6\]](#)

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**Dexmedetomidine (IN premed)**

IN

DOSE 1–3 µg/kg (max 200 µg)

*100 µg/mL solution · onset ~30 min*References: [\[25\]](#), [\[26\]](#)

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**Thiopental**

IV

≤1 mo 3–4 mg/kg

2–12 mo 5–8 mg/kg

≥1 y 5–6 mg/kg

*then 1 mg/kg PRN · titrate to effect*References: [\[27\]](#)

# Painkillers

---

## Paracetamol

IV

$\leq 10$  kg 7.5 mg/kg

$> 10$  kg 15 mg/kg

CAP max 1 g

*interval  $\geq 4$  h*

References: [\[28\]](#), [\[14\]](#), [\[29\]](#)

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## Ketorolac

IV

DOSE 0.5 mg/kg (max 30 mg)

**Contraindicated  $< 6$  months**

References: [\[30\]](#), [\[14\]](#)

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## Morphine

IV

$\leq 1$  mo 0.03–0.05 mg/kg

$\geq 2$  mo 0.1 mg/kg

References: [\[31\]](#), [\[14\]](#)

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## Fentanyl (analgesia)

IV

$\leq 1$  mo 0.5–1  $\mu\text{g/kg}$  — *neonate: reduce · slow injection · watch apnoea*

$\geq 2$  mo 0.5–1  $\mu\text{g/kg}$

References: [\[14\]](#), [\[32\]](#), [\[18\]](#)

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## Metamizole

IV

$\leq 11$  mo 5 mg/kg — *reduced  $< 1$  y — immature metabolism*

$\geq 1$  y 10–20 mg/kg

CAP max 1 g

**Contraindicated  $< 3$  months**

References: [\[33\]](#), [\[34\]](#)

---

## Ibuprofen (IV)

IV

DOSE 10 mg/kg (max 400 mg)

**Contraindicated <3 months**

*not <5 kg · infuse ≥10 min · PDA use is separate*

References: [\[35\]](#), [\[14\]](#), [\[36\]](#), [\[37\]](#), [\[38\]](#)

---

## Ibuprofen (oral)

PO

DOSE 5–10 mg/kg (max 400 mg)

**Contraindicated <3 months**

*not <5 kg*

References: [\[35\]](#), [\[14\]](#), [\[36\]](#)

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## Tramadol

IV/PO

DOSE 1–2 mg/kg (max 100 mg)

**Contraindicated <12 months**

*q6h · respiratory depression caution*

References: [\[39\]](#), [\[40\]](#)

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## Diclofenac (PO/PR)

PO/PR

DOSE 0.3–1 mg/kg

**Contraindicated <12 months**

*q8–12h*

References: [\[41\]](#), [\[40\]](#)

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## Paracetamol (oral)

PO

DOSE 15 mg/kg (max 1 g)

*q4–6h*

References: [\[40\]](#)

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# Emergency Drugs

---

## Adrenaline (cardiac arrest)

IV/IO

DOSE 10 µg/kg (max 1 mg)

*repeat every 3–5 min · max 1 mg/dose*

References: [\[1\]](#), [\[3\]](#)

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## Adrenaline (anaphylaxis IM)

IM

DOSE 10 µg/kg (max 500 µg)

*repeat every 5 min PRN · pens: 150 µg 1–5 y · 300 µg 6–12 y · 500 µg >12 y*

References: [\[1\]](#), [\[4\]](#)

---

## Adrenaline (anaphylaxis IV)

IV

DOSE 1 µg/kg

*titrate to response · IM adrenaline is first-line*

References: [\[1\]](#)

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## Atropine

IV

DOSE 20 µg/kg (max 500 µg)

References: [\[1\]](#), [\[4\]](#), [\[5\]](#)

---

## Ephedrine

IV

≤6 mo 0.6–1.2 mg/kg — ~10× older-child dose — indirect agent, immature sympathetics

≥7 mo 0.1–0.3 mg/kg

*ampoule 30 mg/mL — dilute*

References: [\[42\]](#), [\[14\]](#), [\[6\]](#)

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## Phenylephrine

IV

DOSE 5–20 µg/kg (max 500 µg)

*upper range for tet spells · hypotension: start low, titrate*

References: [\[14\]](#), [\[6\]](#)

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## Lidocaine

IV

DOSE 1 mg/kg (max 100 mg)

*antiarrhythmic — refractory VF / pulseless VT*

References: [\[1\]](#), [\[3\]](#)

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## Glucose 10%

IV/IO

DOSE 2 mL/kg

*hypoglycaemia · recheck glucose after*

References: [\[1\]](#), [\[43\]](#)

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## Calcium gluconate 10%

IV/IO

DOSE 0.5 mL/kg (max 20 mL)

*hypocalcaemia · hyperkalaemia (ECG changes; not in arrest) · CCB overdose · slow IV*

References: [\[1\]](#), [\[43\]](#)

---

## Sodium bicarbonate 8.4%

IV/IO

DOSE 1 mmol/kg

*1 mL/kg of 8.4% · TCA toxicity / severe acidosis · NOT for hyperkalaemic arrest*

References: [\[1\]](#), [\[43\]](#)

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## Dantrolene

IV

DOSE 2.5 mg/kg

*reconstitute vials in 60 mL water NOW · repeat 1 mg/kg every 5 min · may exceed 10 mg/kg*

References: [\[44\]](#), [\[45\]](#)

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**Midazolam (buccal, status epilepticus)**

BUCCAL

 $\leq 11$  mo 2.5 mg

12 mo – 4 y 5 mg

5–9 y 7.5 mg

 $\geq 10$  y 10 mg**Contraindicated <3 months***seizure  $\geq 5$  min · 3–6 mo: hospital only · repeat once after 5–10 min*References: [\[46\]](#), [\[1\]](#)

---

**Midazolam (IV, status epilepticus)**

IV/IO

DOSE 0.15 mg/kg (max 10 mg)

*repeat once · off-label use*References: [\[47\]](#), [\[48\]](#), [\[1\]](#)

---

**Hydrocortisone (adrenal crisis)**

IV

 $\leq 1$  mo 4 mg/kg $\geq 2$  mo 2 mg/kg

CAP max 100 mg

*then 1 mg/kg (max 50) q6h · saline 10 mL/kg bolus · glucose if low*References: [\[49\]](#), [\[1\]](#)

---

**Insulin + glucose (hyperkalaemia)**

IV

DOSE 0.1 units/kg (max 10 units)

*WITH glucose 10% 5 mL/kg — never insulin alone · check glucose & K<sup>+</sup> q15min*References: [\[50\]](#), [\[1\]](#)

---

**Salbutamol (nebulized, hyperkalaemia)**

NEBULIZED

VALUE 2.5–5 mg

*2.5 mg  $\leq 25$  kg · 5 mg  $> 25$  kg · repeat  $\leq 5\times$  · with insulin–glucose*References: [\[50\]](#), [\[1\]](#)

# Fluids and Blood

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## Fluid Bolus

IV

DOSE 10 mL/kg (max 500 mL)

*balanced crystalloid · cardiogenic shock: 5 mL/kg*

References: [\[1\]](#)

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## Maintenance

RULE Holliday-Segar 4-2-1: 4 mL/kg/h (first 10 kg) · 2 mL/kg/h (next 10 kg) · 1 mL/kg/h thereafter

*4-2-1 (Holliday-Segar) · isotonic + glucose ± K+ · <1 mo: neonatal regimen*

References: [\[51\]](#), [\[52\]](#)

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## Total Blood Volume

≤1 mo 85 mL/kg

2 mo – 2 y 75 mL/kg

≥3 y 70 mL/kg

References: [\[6\]](#)

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## Tranexamic Acid

IV

DOSE 15 mg/kg (max 1 g)

*loading dose over 10 min · then 2 mg/kg/h infusion*

References: [\[53\]](#), [\[54\]](#), [\[55\]](#)

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## Max allowable blood loss

VALUE  $EBV \times (Hct - \text{min Hct}) / Hct$

*estimate, not a transfusion trigger · EBV ~70–100 mL/kg*

References: [\[56\]](#), [\[57\]](#)

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# Equipment

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## ETT

RULE Cuffed:  $3.5 + \text{age}/4$ , rounded to  $\pm 0.5$

<1 MO Neonatal weight table ( $\leq 4$  kg, uncuffed):

$\leq 1.5$  kg    size 2.5

$\leq 3$  kg    size 3.0

$\leq 4$  kg    size 3.5

INFANT age table (cuffed):

$\leq 6$  mo    size 3.5

7–12 mo    size 4.0

*cuffed sizes — uncuffed add  $0.5 (\geq 1 \text{ y}) \cdot \text{neonatal table: uncuffed sizes}$*

References: [\[58\]](#), [\[59\]](#)

---

## ETT insertion depth

RULE  $6 + \text{weight} (<1 \text{ mo}) \cdot \text{weight}/2 + 8 (<1 \text{ y}) \cdot \text{age}/2 + 12 (\geq 1 \text{ y})$  (cm)

*nasal:  $+1 \text{ cm} <1 \text{ y} \cdot +2-3 \text{ cm} \geq 1 \text{ y}$*

References: [\[60\]](#), [\[61\]](#), [\[43\]](#)

---

## iGel LMA

IGEL SIZING weight bands (inclusive on both bounds):

2–5 kg    size 1

5–12 kg    size 1.5

10–25 kg    size 2

25–35 kg    size 2.5

30–60 kg    size 3

50–90 kg    size 4

90– $\infty$  kg    size 5

References: [\[62\]](#), [\[63\]](#)

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## AuraGain LMA

AURAGAIN SIZING weight bands (inclusive on both bounds):

|           |          |
|-----------|----------|
| 2–5 kg    | size 1   |
| 5–10 kg   | size 1.5 |
| 10–20 kg  | size 2   |
| 20–30 kg  | size 2.5 |
| 30–50 kg  | size 3   |
| 50–70 kg  | size 4   |
| 70–100 kg | size 5   |
| 100–∞ kg  | size 6   |

References: [\[64\]](#)

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## Laryngoscope blade

|       |             |
|-------|-------------|
| ≤1 y  | Miller 0–1  |
| 2–7 y | Macintosh 2 |
| ≥8 y  | Macintosh 3 |

*Miller lifts the epiglottis · Macintosh 1 equally effective*

References: [\[65\]](#), [\[6\]](#)

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## Face mask

|         |        |
|---------|--------|
| ≤1 mo   | Size 0 |
| 2–12 mo | Size 1 |
| 1–3 y   | Size 2 |
| 4–8 y   | Size 3 |
| 9–14 y  | Size 4 |
| ≥15 y   | Size 5 |

*cover nose to chin cleft — anatomical fit over size*

References: [\[6\]](#), [\[66\]](#)

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## Reservoir bag

≤6 mo      0.5 L

7 mo – 6 y    1 L

≥7 y      2 L

References: [\[67\]](#), [\[68\]](#)

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## Suction catheter

≤1 mo      6 Fr

2 mo – 1 y    8 Fr

2–8 y      10 Fr

≥9 y      12–14 Fr

References: [\[43\]](#), [\[6\]](#)

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## NG/OG tube

≤1 mo      6 Fr

2 mo – 1 y    8 Fr

2–8 y      10 Fr

≥9 y      12–14 Fr

*larger sizes for adolescent decompression*

References: [\[43\]](#), [\[6\]](#)

---

## OPA / NPA sizing

VALUE by landmark

*OPA: incisors → angle of mandible · NPA: nostril → tragus; not in basal-skull fracture*

References: [\[48\]](#), [\[69\]](#), [\[70\]](#)

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# Respiratory

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## Salbutamol (inhaled)

INHALED

≤5 y 2–10 puffs

≥6 y 4–10 puffs

*100 µg/puff · repeat incrementally as needed*

References: [\[71\]](#), [\[72\]](#)

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## Salbutamol (nebulized)

NEBULIZED

≤5 y 2.5 mg

≥6 y 5 mg

References: [\[71\]](#), [\[72\]](#)

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## Ipratropium (nebulized)

NEBULIZED

≤11 y 250 µg

≥12 y 500 µg

References: [\[14\]](#)

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## Magnesium Sulphate

IV

DOSE 25–50 mg/kg (max 2 g)

*severe asthma · infuse over 20 min*

References: [\[73\]](#), [\[14\]](#), [\[1\]](#)

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## Clemastine

IV

DOSE 25 µg/kg (max 2 mg)

**Contraindicated <12 months**

*second-line, after adrenaline*

References: [\[74\]](#), [\[75\]](#), [\[76\]](#)

---

**Adrenaline (nebulized)**

NEBULIZED

DOSE 0.5 mg/kg (max 5 mg)

*croup / stridor · 1:1000 (1 mg/mL)*References: [\[77\]](#), [\[14\]](#)

---

**Propofol (laryngospasm)**

IV

DOSE 0.5 mg/kg

*give in increments*References: [\[78\]](#)

---

**Succinylcholine (laryngospasm)**

IV

DOSE 0.1–0.5 mg/kg

*lower end may spare spontaneous ventilation · avoid in MH / hyperkalaemia risk*References: [\[79\]](#), [\[78\]](#)

---

**Succinylcholine (IM)**

IM

DOSE 4 mg/kg (max 200 mg)

*laryngospasm / RSI without IV access · avoid in MH / hyperkalaemia risk*References: [\[79\]](#), [\[78\]](#)

---

**Rocuronium (laryngospasm)**

IV

DOSE 0.6–1.2 mg/kg

*when succinylcholine contraindicated (MH) — then secure the airway*References: [\[16\]](#), [\[15\]](#)

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**Salbutamol (IV, severe asthma)**

IV

DOSE 15 µg/kg (max 250 µg)

*bolus over 10 min · infusion 1–2 µg/kg/min (>2: PICU) · monitor K+ · off-label <12 y*References: [\[71\]](#), [\[80\]](#)

---

**Prednisolone (PO, asthma)**

PO

DOSE 1–2 mg/kg (max 40 mg)

*within the first hour · oral = IV efficacy*References: [\[80\]](#), [\[71\]](#), [\[48\]](#)

---

**Hydrocortisone (IV, asthma)**

IV

DOSE 4 mg/kg

*q4h — if oral not tolerated*References: [\[71\]](#)

# Reversal Agents

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## Naloxone (titrated)

IV

DOSE 10 µg/kg

*titrate to respiration · repeat q2–3 min*

References: [\[14\]](#), [\[1\]](#)

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## Naloxone (overdose)

IV/IO/IM

DOSE 0.1 mg/kg (max 2 mg)

References: [\[1\]](#), [\[14\]](#)

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## Flumazenil

IV

DOSE 10 µg/kg (max 200 µg)

*repeat ≤4× · total max 50 µg/kg (up to 1 mg) · licensed ≥1 y*

References: [\[14\]](#)

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## Sugammadex

IV

DOSE 2–4 mg/kg

*upper dose for deep block*

References: [\[81\]](#), [\[82\]](#)

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## Sugammadex (profound block)

IV

DOSE 16 mg/kg

*CICO rescue — reverses full-dose rocuronium · off-label in children*

References: [\[83\]](#), [\[84\]](#)

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## Neostigmine

IV

DOSE 0.04–0.07 mg/kg (max 5 mg)

*with atropine · slow over 1–2 min, after first recovery signs*

References: [\[14\]](#), [\[6\]](#)

---

**Atropine (with neostigmine)**

IV

DOSE 20 µg/kg (max 500 µg)

*with or just before neostigmine*References: [\[14\]](#), [\[6\]](#)

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**Neostigmine + glycopyrronium**

IV

DOSE 0.02 mL/kg (max 2 mL)

References: [\[85\]](#)



# Regional Anesthesia

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## Lidocaine (perineural)

PERINEURAL

≤5 mo    maximum 3.5 mg/kg

≥6 mo    maximum 5 mg/kg

*plain solution — no adrenaline*

References: [\[6\]](#), [\[14\]](#)

---

## Bupivacaine (perineural)

PERINEURAL

≤5 mo    maximum 1.5–2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo    maximum 2–2.5 mg/kg — *infusion: 0.4 mg/kg/h*

References: [\[86\]](#), [\[87\]](#)

---

## Ropivacaine (perineural)

PERINEURAL

≤5 mo    maximum 2 mg/kg — *infusion: 0.2 mg/kg/h*

≥6 mo    maximum 3 mg/kg — *infusion: 0.4 mg/kg/h*

References: [\[88\]](#), [\[89\]](#), [\[87\]](#)

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## Intralipid 20% bolus

IV

DOSE    1.5 mL/kg

*LAST rescue · max total 12 mL/kg*

References: [\[86\]](#), [\[90\]](#)

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## Intralipid 20% infusion

IV

DOSE    0.25 mL/kg/min

*LAST rescue – continue until stable*

References: [\[86\]](#), [\[90\]](#)

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## Levobupivacaine (max)

PERINEURAL

DOSE maximum 1.3 mg/kg (max 150 mg)

**Contraindicated <6 months**

References: [\[91\]](#)

# Weight Estimation

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Used when the user does not supply a measured weight. <1 month and >12 years always require user input.

## 1–12 mo (infant)

FORMULA  $\text{weight (kg)} = (0.5 \times \text{months}) + 4$

References: [\[43\]](#)

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## 1–5 y (young child)

FORMULA  $\text{weight (kg)} = (2 \times \text{age}) + 8$

References: [\[43\]](#)

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## 6–12 y (school-age)

FORMULA  $\text{weight (kg)} = (3 \times \text{age}) + 7$

References: [\[92\]](#), [\[43\]](#)

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